

Last time

n-simplex $\Delta(S)$ convex hull in $\mathbb{R}^N$ of a set S
of size $n + 1$ in general pos

subsimplex $\Delta(S')$ for some $S' \text{ sub } S$

boundary $\partial\Delta(S)$ union of proper subsimplices

interior $\text{Int}(\Delta(S))$ $\Delta(S) - \partial\Delta(S)$

simplicial complex $K = \{\Delta(S)\}_{S \in I}$ s.t.
1) the $\text{Int}(\Delta(S))$ partition
 $\bigcup_{S \in I} \Delta(S)$
2) if $S \in I$ and $S' \text{ sub } S$,
then $S' \in I$ too

$|K| = \bigcup_{S \in I} \Delta(S)$ is the realization of K

today: instead of $\Delta(\{v_0, v_1, \dots, v_k\})$
we will write $[v_0, v_1, \dots, v_k]$ [for convenience]

Ex pick a, b, c in $\mathbb{R}^3$
[which are simplicial complexes?]
 $\{[a], [b], [c], [a, b], [b, c]\}$
 $\{[a], [b], [c], [a, b]\}$
 $\{[a], [c], [a, b]\}$

Ex pick $p < q < r$ in $\mathbb{R}$
[which are simplicial complexes?]
 $\{[p], [q], [r], [p, q], [p, r]\}$
 $\{[p], [q], [r], [p, q], [q, r]\}$
 $\{[p], [q], [r], [p, r]\}$

Euler characteristic of K :

$$\chi(K) = e_0(K) - e_1(K) + e_2(K) - \dots$$

where $e_n(K) = \#$ of n -simplices in K

Ex $\{[p], [q], [r], [p, q], [q, r]\}$
has Euler char $3 - 2 = 1$

cf. Problems 67–68

Ex simplicial tetrahedron has Euler char
 $4 - 6 + 4 = 2$ [cf. Euler's formula]

Thm if $|K_1| \sim |K_2|$, then $\chi(K_1) = \chi(K_2)$

Q is the converse true?

A no! tetrahedron vs $\{0, 1\}$

(Crossley §9.1) goal: find an improvement of χ
that can distinguish
tetrahedron vs $\{0, 1\}$

idea #1 instead of a single number χ ,
use the sequence $e_0, e_1, e_2, \dots$

e_i 's not invariant under homotopy equivalence!

idea #2 modify the e_i 's to get an invariant
sequence $b_0, b_1, b_2, \dots$

Ex $K_1 = \{[a], [b], [c], [a, b]\}$
 $K_2 = \{[a], [c]\}$

then $|K_1| \sim |K_2|$
 $(e_i(K_1))_i = (3, 1, 0, 0, \dots)$
 $(e_i(K_2))_i = (2, 0, 0, 0, \dots)$

notice: $\partial\Delta(\{a, b\}) = \Delta(\{a\}) \cup \Delta(\{b\})$

idea #3 use ∂ to encode how the segment
 between a and b contracts down to a

Chains with $\mathbb{Z}/2\mathbb{Z}$ Coefficients

Df let $S_n(K)$ be the set of n-simplices in K
 let $C_n(K)$ be the power set of $S_n(K)$

we can view $C_n(K)$ as an abelian group!

write any elt of $C_n(K)$ as a formal sum: e.g.,
 if $c = \{\Delta_1, \Delta_2, \Delta_3\}$ in $C_n(K)$
 then as a formal sum, $c = \Delta_1 + \Delta_2 + \Delta_3$

the group law is addition mod 2: e.g.,
 $(\Delta_1 + \Delta_2 + \Delta_3) + (\Delta_3 + \Delta_4)$
 $= \Delta_1 + \Delta_2 + \Delta_4$

Rem as a formal sum, $\emptyset$ becomes 0

Rem since $c + c = 0$ for all c in $C_n(K)$
 can view $C_n(K)$ as a $\mathbb{Z}/2\mathbb{Z}$ -vector space

elts of $C_n(K)$ are called $\mathbb{Z}/2\mathbb{Z}$ simplicial chains

any n -simplex Δ in K can now be viewed as a simplicial chain Δ in $C_n(K)$

Df the boundary chain of Δ is

$$\partial\Delta = \sum_{\{\text{faces } \Delta'\}} \Delta' \text{ in } C_{\{n-1\}}(K)$$

[recall: faces are maximal proper subsimplices]

Ex $\partial[a, b, c] = [a, b] + [a, c] + [b, c]$

the boundary map $\partial_n : C_n(K)$ to $C_{\{n-1\}}(K)$ is defined as follows [“by linearity”]:

$$\begin{aligned} \text{if } c &= \sum_i \Delta_i \\ \text{then } \partial c &= \sum_i \partial\Delta_i \end{aligned}$$

Df

$$\begin{aligned} \text{let } Z_0(K) &= C_0(K) \\ \text{let } Z_n(K) &= \ker(\partial_n) && \text{for } n > 0 \\ \text{let } B_n(K) &= \text{im}(\partial_{\{n+1\}}) \end{aligned}$$

elts of $Z_n(K)$ are called $\mathbb{Z}/2\mathbb{Z}$ cycles

elts of $B_n(K)$ are called $\mathbb{Z}/2\mathbb{Z}$ boundaries

Ex

$$\begin{aligned} K &= \{[a], [b], [c], [a, b]\} \\ C_0 &\text{ is free abelian on } [a], [b], [c] \\ C_1 &\text{ is free abelian on } [a, b] \end{aligned}$$

$$\begin{aligned} \partial : C_1 &\text{ to } C_0 \text{ is def by } \partial[a, b] = [a] + [b] \\ Z_0 &\text{ is free abelian on } [a], [b], [c] \\ Z_1 &\text{ is trivial} \\ B_0 &\text{ is free on } [a] + [b] \\ B_1 &\text{ is trivial} \end{aligned}$$

Df the mod 2 simplicial homology of K is

$$H_*(K) = \bigoplus_n H_n(K)$$

where the deg- n part is $H_n(K) = Z_n(K)/B_n(K)$

the n th mod 2 Betti number of K is

$$b_n = \dim_{\{Z/2Z\}} H_n(K)$$

i.e., size of any minimal generating set for $H_n(K)$

Thm $(b_i(K))_i$ only depends on
the homotopy equivalence class of $|K|$

Thm $b_0 - b_1 + b_2 - \dots = \chi(K)$